

Apply Single Quotes, Preserve Literal Content, Ignore Variable Expansion, Store Quoted Strings, Validate Parsing Results, Test Edge Cases

Code:

```
geya@LenovoL480: ~/OS/Skil × + v
GNU nano 8.7.1
#include <stdio.h>
#include <string.h>

int main() {
    char input[100];
    char output[100];
    int i = 0, j = 0;
    int single_quote = 0;

    printf("Enter text: ");
    fgets(input, sizeof(input), stdin);

    while (input[i] != '\0') {
        if (input[i] == '\\') {
            single_quote = !single_quote;
        }
        else {
            output[j++] = input[i];
        }

        i++;
    }

    output[j] = '\0';
```

Output:

```
geya@LenovoL480:~/OS/Skill/Week 3$ ./task1
Enter text: Hello
Parsed text: Hello

Parsing successful
geya@LenovoL480:~/OS/Skill/Week 3$ ./task1
Enter text: 'Hi"
Parsed text: Hi"

Error: Unclosed single quote
geya@LenovoL480:~/OS/Skill/Week 3$ ./task1
Enter text: 'HI'
Parsed text: HI

Parsing successful
geya@LenovoL480:~/OS/Skill/Week 3$ |
```